Pass@k Hides Differential Collapse: A Controlled Study of Post-Training in Medical Diagnosis

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Abstract

A clinical diagnosis is a ranked differential, not a single answer, yet language models are mostly judged by whether their top answer is correct. We study what post-training does to the rest of the differential. We compare base models with their post-trained descendants under a controlled design: two base/instruct pairs (Qwen2.5-7B, Qwen3-8B), and three public medical models fine-tuned from the same parent with reinforcement learning, SFT with PPO, or distillation. We evaluate them with 32 samples per case on DDXPlus, whose patients come with a reference differential, and on rare-disease cases from Phenopacket Store. General post-training raises first-answer accuracy by 17 to 27 points but lowers coverage at larger sample budgets: base models are 10 to 15 points better at pass@32. The post-trained models' sampled differentials collapse onto one or two diagnoses (answer entropy 0.18 vs. 1.31 for the base at T=1, against 1.87 for the reference), even though their top-1 calibration improves. Medical RL applied after instruction tuning improves pass@k at every k but does not undo the collapse. On rare disease, medical post-training raises stated confidence from 0.51 to 0.66–0.69 while accuracy stays below 2%. Single-answer metrics, including pass@k at small k, therefore rate the post-trained models as strictly better while missing a large loss of diagnostic coverage. Per-sample RL rewards inherit this narrowing; a group-level reward that scores the spread of sampled answers removes it, raising accuracy and diversity together in GRPO fine-tuning. [TODO: update with 8B result]

1 Introduction

Clinicians reason over a differential diagnosis, a ranked set of conditions consistent with the presentation, and the conditions they keep in view determine which tests are ordered and which dangerous possibilities are ruled out. Language-model evaluations in medicine nonetheless score a single answer: multiple-choice accuracy, top-1 diagnosis, or at most pass@k for a small k. Post-training, whether general instruction tuning and RLHF or domain-specific medical reinforcement learning and distillation, is reported to improve these scores [1, 8, 14]. Work on reinforcement learning with verifiable rewards outside medicine suggests a cost: RL often sharpens a model's output distribution, so post-trained models win at pass@1 while their base models solve more problems given many samples [16, 19, 20]. Whether a similar trade-off holds for diagnosis, and whether it shows up in the differential rather than only in pass@k, has not been measured.

We measure it directly. Our design holds the pretrained model fixed and varies only the post-training: (i) two base/instruct pairs, Qwen2.5-7B and Qwen3-8B, which isolate general post-training; (ii) three public medical models fine-tuned from the same parent, Qwen2.5-7B-Instruct: AlphaMed (rule-based RL), HuatuoGPT-o1 (SFT then PPO), and m1 (SFT distillation). We draw 32 samples per case and evaluate each model's sampled differentials against a reference distribution

on DDXPlus [7], which ships a differential for every synthetic patient, and against an ontology on rare-disease cases from Phenopacket Store [5].

Findings.

1. **General post-training trades coverage for top-1 accuracy.** In both families the post-trained model is far better on its first answer (+17 to +27 points) but worse at larger sample budgets. Its pass@k curve crosses the base model's at k≈4–8, and at k=32 the base model leads by 10–15 points. All intervals exclude zero (§4.1).

2. **The differential collapses, and single-answer metrics do not show it.** Post-trained models give 1.3–2.6 distinct top answers per 32 samples, against 3.8–8.3 for their base models. Their answer entropy is 0.11–0.49, against 0.78–1.59 for base models and 1.87 for the reference differential. This holds even when sampling at T=1. Their top-1 calibration is *better*, so on every single-answer metric they look strictly superior (§4.2).

3. **Medical RL on an instruction-tuned model adds accuracy without undoing the collapse.** AlphaMed improves on its parent at every k (+3.9 at k=1, +8.4 at k=32), so we see no sharpening tax at the medical stage. Its answer diversity matches its parent's, which is already collapsed (§4.3).

4. **On rare disease, medical post-training increases overconfidence.** With accuracy below 2% for every model, mean stated confidence rises from 0.51 for the parent to 0.66–0.69 for its RL-trained descendants (§4.4).

5. **A group-level reward avoids the collapse.** Rewarding each GRPO sample for its contribution to the match between the group's sampled answers and the reference differential raises accuracy as much as standard RLVR, while preserving and then widening the differential; per-sample distributional rewards do not (§5).

2 Related work

What RL does to a model's output distribution. Yue et al. [20] find that RL with verifiable rewards improves pass@1 on math and code while base models win at large k. Wu et al. [19] argue that RLVR mostly reweights solutions already in the base model's support. Oh et al. [16] document a "sharpening tax" across 14 base/post-trained pairs on agentic benchmarks. In contrast, Cheng et al. [3] report that RL expands the boundary on domains under-represented in pretraining. In medicine, the only pass@k analysis we know of studies vision-language models and finds that RL mainly sharpens [9]. We give the first text-only medical analysis, and we measure the differential directly rather than only pass@k.

Diversity loss from alignment. Instruction tuning and RLHF reduce output diversity [11, 18]. Related work explains this as typicality bias [21] and documents cross-model homogeneity [10]. We quantify this loss against a clinical reference distribution.

Medical reasoning models and calibration. Medical post-training recipes include verifier-guided SFT and PPO [1], minimalist rule-based RL [14], and test-time-scaling distillation [8]. Reward models and RL tend to favor confident answers [4, 12], and reasoning models are often overconfident [15]. We show that this overconfidence appears in public medical descendants of one parent, on rare disease.

Model	Parent	Post-training	Role
Qwen2.5-7B	n/a	none (pretrained)	base
Qwen2.5-7B-Instruct	Qwen2.5-7B	general SFT + RLHF	instruct; medical parent
AlphaMed-7B [14]	Qwen2.5-7B-Instruct	rule-based RL	medical
HuatuoGPT-o1-7B [1]	Qwen2.5-7B-Instruct	SFT + PPO (verifier)	medical
m1-7B-23K [8]	Qwen2.5-7B-Instruct	SFT distillation	medical
Qwen3-8B-Base	n/a	none (pretrained)	base
Qwen3-8B	Qwen3-8B-Base	general post-training	instruct

Table 1: Models studied. Each pair or family shares its pretrained weights, so differences reflect post-training.

Differential diagnosis benchmarks and distributional RL. DDXPlus [7] was introduced for automatic diagnosis systems trained on differentials [6]. Recent LLM evaluations on it score hierarchical top-k accuracy [13]. Puri et al. [17] train models with GRPO to output several answers per forward pass, rewarding coverage of a gold *set* and set-level calibration, and evaluate on DDXPlus. We instead measure collapse against the reference *distribution* across public post-trained model families. Our group-level reward instead operates on the distribution of independently sampled answers (§5).

3 Experimental setup

Tasks. DDXPlus [7]: synthetic patients produced by a rule-based medical knowledge base. Each patient has a true pathology and a reference differential, a probability distribution over the 49 conditions. We use 980 test patients, 20 per condition. Each patient's coded symptoms and history are rendered as a short vignette; we fix known mistranslations in the English evidence descriptions. The prompt lists all 49 conditions and asks for the five most likely, with probabilities, as JSON. **Phenopacket Store** [5] (v0.1.27): published rare-disease cases annotated with HPO phenotypes and an OMIM diagnosis. We use a disease-balanced set of 500 cases covering 418 diseases. Each case is a vignette of present and absent phenotypes, and the model is asked for an open-ended differential. Answers are graded by mapping names to Mondo; we credit an exact match, and separately a "related" match to a specific ancestor group or subtype.

Models. Table 1. All checkpoints are public. For the medical models we verified the parent model from each released configuration.

Sampling. We draw n=32 samples per case with vLLM, at T=0.7, top-p 0.95, and at T=1.0, top-p 1.0 (sampling from the model's own distribution). All models use their chat template and the same prompt. Completions are capped at 4,096 tokens (8,192 for m1, whose long reasoning otherwise truncates), and generation stops at the end-of-turn token. We take the last valid JSON list in each completion. Unparsable completions count as incorrect; this affects mainly base models (11–19%), so base-model pass@k is a lower bound.

Metrics. *pass@k*: the unbiased estimator of Chen et al. [2] on the top-1 answer of each sample. *Distinct* and *entropy*: the number of distinct top-1 diagnoses across a case's 32 samples, and the entropy of their empirical distribution, averaged over cases. On DDXPlus, TV: total-variation distance between this empirical distribution and the reference differential. *Coverage@k*: the expected

Pair	Model	pass@1	pass@32	Distinct	Entropy	TV↓	Cov@32	ECE↓
Qwen3-8B, T=1	base	0.282	**0.789**	**6.60**	**1.31**	**0.712**	**0.407**	0.214
	post-trained	**0.556**	0.643	1.53	0.18	0.783	0.236	**0.061**
Qwen2.5-7B, T=1	base	0.235	**0.757**	**8.32**	**1.59**	**0.712**	**0.426**	0.309
	instruct	**0.457**	0.657	2.63	0.49	0.761	0.290	**0.137**
Qwen3-8B, T=0.7	base	0.388	**0.730**	**3.76**	**0.78**	**0.750**	**0.322**	0.121
	post-trained	**0.558**	0.607	1.27	0.11	0.793	0.217	**0.063**

Table 2: DDXPlus, 32 samples per case (980 cases; 811 for Qwen2.5 at T=1, [TODO: complete]). The reference differential has mean entropy 1.87. Distinct and entropy refer to the top-1 answers across samples; Cov@32 is the share of reference probability mass covered by those answers. ECE is for the stated top-1 probability.

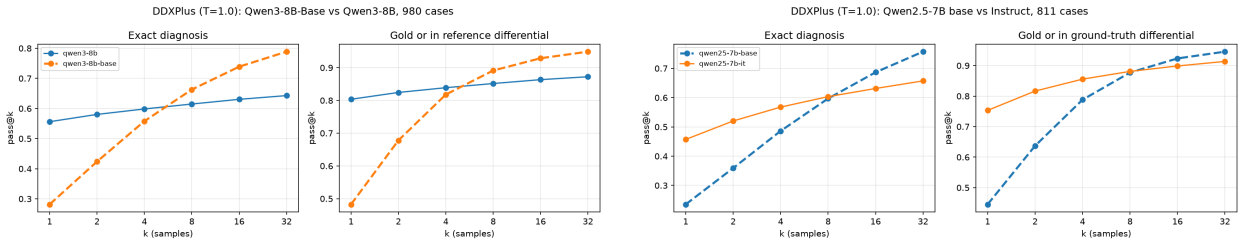


Figure 1: pass@k on DDXPlus at T=1. Left: Qwen3-8B-Base vs. Qwen3-8B. Right: Qwen2.5-7B base vs. Instruct. In each, the post-trained model wins at small k and loses at large k. Right panels of each plot count a sample as correct if its answer is the gold condition or appears in the reference differential. [TODO: redraw as one paper-style figure with CIs]

share of the reference differential's probability mass covered by the distinct top-1 answers in k random samples. *Calibration*: expected calibration error (10 bins) of the stated top-1 probability against top-1 correctness. Differences come with 95% paired-bootstrap intervals over cases (2,000 resamples).

4 Results

4.1 General post-training trades coverage for top-1 accuracy

Table 2 and Figure 1 compare each base model with its post-trained counterpart on DDXPlus. In both families the post-trained model is much better on its first answer: +27.5 points for Qwen3 and +22.2 for Qwen2.5 at T=1 (intervals $[+25.3, +29.6]$ and $[+20.1, +24.4]$). The advantage shrinks with k and reverses: the curves cross at k≈4–8, and at k=32 the base model leads by 14.6 points (Qwen3, $[+12.2, +17.1]$) and 10.0 points (Qwen2.5, $[+7.1, +12.8]$). The pattern is the same at T=0.7 for Qwen3, crossing at k=4 with a base lead of 12.2 points at k=32 ($[+9.9, +14.9]$). Unparsable base-model samples count as wrong, so these gaps are lower bounds. [TODO: Qwen2.5 base pair on all 980 cases (currently 811); Qwen2.5 base at T=0.7.]

4.2 The differential collapses, while top-1 calibration improves

The gap at large k reflects a collapsed differential. Across 32 samples, post-trained models name 1.3–2.6 distinct diagnoses per case, against 3.8–8.3 for their base models. Their answer entropy

Comparison	Considered@32 (base → post-trained) Severe (1–2)	Moderate (3)	Mild (4–5)
Qwen3-8B, T=1	0.45 → 0.27	0.50 → 0.32	0.45 → 0.20
	−18.4 [−20.8, −15.9]	−18.1 [−20.7, −15.7]	−24.2 [−27.0, −21.4]
Qwen2.5-7B, T=1	0.52 → 0.34	0.50 → 0.38	0.39 → 0.23
	−18.2 [−21.4, −15.2]	−12.3 [−15.2, −9.4]	−15.4 [−18.2, −12.5]
Qwen3-8B, T=0.7	0.34 → 0.24	0.44 → 0.29	0.31 → 0.18
	−9.8 [−11.8, −7.9]	−15.8 [−18.2, −13.5]	−13.4 [−15.8, −11.1]
Instruct → AlphaMed, T=0.7	0.28 → 0.32	0.32 → 0.34	0.17 → 0.21
	+4.3 [+2.0, +6.6]	+2.2 [+0.3, +4.1]	+3.9 [+2.4, +5.5]

Table 3: Share of (case, condition) pairs with reference probability ≥ 0.1 in which the condition is a top-1 answer in at least one of 32 samples, by DDXPlus severity. Second rows give differences in points with 95% case-bootstrap intervals. Post-training lowers consideration in every tier, including the can't-miss tier; medical RL slightly raises it. (Qwen2.5 at T=1: 811 cases, [TODO: complete].)

is 0.11–0.49, against 0.78–1.59 for base models and 1.87 for the reference differential (Table 2). Sampling at T=1 widens the base models' answers but barely changes the post-trained models' (Qwen3 entropy 0.11 at T=0.7, 0.18 at T=1), so the collapse lives in the model's own distribution rather than in the decoding settings. As a result, post-trained models cover less of the reference differential: 0.24 vs. 0.41 of its probability mass at k=32 for Qwen3, and 0.29 vs. 0.43 for Qwen2.5. Their top-1 calibration, however, is markedly better (ECE 0.06 vs. 0.21 and 0.14 vs. 0.31). On every single-answer criterion (accuracy, small-k pass@k, calibration), the post-trained model therefore looks strictly better, and the loss of diagnostic breadth is invisible.

Which diagnoses drop out? For each case and each condition that the reference differential weights at least 0.1, we ask whether the model names that condition as its top answer in at least one of its 32 samples. We call this *considering* the condition. We group the 49 conditions by DDXPlus severity (1 = most severe). Post-training lowers consideration in every tier (Table 3), and the severe "can't-miss" tier is not spared. For the 17 severity-1–2 conditions (e.g. NSTEMI/STEMI, acute pulmonary edema, anaphylaxis, pulmonary embolism, unstable angina), consideration falls from 0.45 to 0.27 for Qwen3 and from 0.52 to 0.34 for Qwen2.5 at T=1. The differences are −18.4 [−20.8, −15.9] and −18.2 [−21.4, −15.2] points. Some diagnoses disappear from the top answer almost entirely. Qwen2.5-7B-Instruct names unstable angina as its top answer in one of 107 eligible cases, where its base model does in 47%. When unstable angina is the true pathology, pass@32 falls from 1.00 to 0.00. Stable angina (0.89 to 0.06) and myocarditis (0.41 to 0.00) follow the same pattern. These conditions are not forgotten: they still appear lower in the listed differential (unstable angina in 50% of eligible cases, against 95% for the base model). The model ranks them but no longer commits to them. At the same time, post-trained models concentrate on a few salient "attractor" diagnoses. Qwen2.5-7B-Instruct gives NSTEMI/STEMI as its top answer in 13.1% of samples, against a reference rate of 4.4% and 6.4% for its base model. It also over-predicts pulmonary embolism (8.2% vs. 3.2%) and pneumonia (8.0% vs. 2.4%). Chest-pain presentations thus collapse onto the most dangerous candidate. That raises its accuracy (NSTEMI pass@1 0.99 vs. 0.44) at the cost of the alternatives, which is a defensible triage bias but not a differential.

4.3 Medical post-training on an instruction-tuned model

Applied after instruction tuning, medical RL does not cost coverage. AlphaMed improves on its parent at every k (Figure 3): +3.9 points at k=1 ([+2.2, +5.5]), widening to +8.4 at k=32 ([+6.0, +10.6]). It reaches at least one correct sample for 47 of 49 conditions, against 39 for its parent. It does not restore breadth, though: distinct answers (1.87 vs. 1.82), entropy (0.29 vs. 0.30), and coverage (0.27 vs. 0.24) stay at the instruction-tuned level, far below the base model. AlphaMed also names off-list conditions more often than its parent, a mild loss of instruction following. By condition, AlphaMed slightly *raises* consideration in every severity tier (+2.2 to +4.3 points; Table 3). It still loses acute pulmonary edema (0.31 to 0.13), and it introduces an attractor of its own: spontaneous pneumothorax as the top answer in 10.3% of samples, against 4.3% for its parent and a reference rate of 1.0%. Medical post-training thus operates on an already collapsed distribution. The collapse comes from the general post-training stage. [TODO: HuatuoGPT-o1 (SFT+PPO) and m1 (SFT distillation) on DDXPlus; on Phenopackets HuatuoGPT is narrower and m1 broader than the parent.]

4.4 Rare disease: overconfidence without accuracy

On 500 Phenopacket Store cases covering 418 diseases, every 7B model is near the floor: pass@1 is 1.3–1.9% and pass@32 is 6–7%, and differences in pass@k are within noise. Stated confidence is not near the floor. The parent model states a mean top-1 probability of 0.51, and its medical descendants state 0.66 (HuatuoGPT-o1) and 0.69 (AlphaMed), while their accuracy stays at 1.9%. ECE accordingly rises from 0.50 to 0.64–0.67. Medical post-training thus increases overconfidence exactly where the model is out of its depth. HuatuoGPT-o1 is also the narrowest model here (9.4 distinct answers per 32 samples, against 11.5 for its parent). Accuracy depends strongly on when the source case report was published, which suggests memorization. Across the Qwen2.5 family, pass@32 is 8.1–9.6% on cases published up to 2022 (n=334), 0.9–1.7% on 2023–24 cases (n=115), and 3.9% on 2025+ cases (n=51). Even the low absolute accuracy is therefore inflated by older, likely-memorized cases. [TODO: m1 with an 8k token budget; Qwen3 pair; full 1,401-case set].

5 Training against the collapse

Can reinforcement learning raise diagnostic accuracy without narrowing the differential? We fine-tune Qwen3-1.7B with GRPO on DDXPlus training patients (9,800 cases, 200 per condition; disjoint from the test split) and evaluate exactly as in §3.

Setup. We train LoRA adapters (r=32) with GRPO for 200 steps. Each step uses 16 prompts and 16 sampled completions per prompt, generated with vLLM on the same GPU. A first sweep with a common aggressive configuration (learning rate 2×10^{-5}, no KL penalty, 8 samples per prompt, per-group reward standardization) was unstable. In three of seven runs, the policy collapsed to an input-independent answer: one condition for 88–100% of all patients, with completions shrinking to about 25 tokens. The starting policy is nearly deterministic (token entropy ≈0.06), so most GRPO groups have zero reward variance, and noisy updates drive the collapse. We therefore use learning rate 5×10^{-6}, KL coefficient 0.04, no reward standardization (as in Puri et al. [17]), and 16 samples per prompt. A monitor tracks the number of distinct top-1 answers per case within each batch. All runs in this configuration were stable.

Rewards. Every recipe includes a format reward: the answer must parse into a list of distinct conditions with valid probabilities. `acc` is standard RLVR: 1 if the top-1 answer is the true pathology. `dist` adds two per-sample distributional terms to `acc`: a Brier score of the stated distribution against the truth, and the overlap of the stated distribution with the reference differential. Both are computed from a *single* completion. They cannot see that the policy gives the same answer in every sample, which is exactly the collapse of §4.2. `gm` adds to `acc` a *group-level* reward over the GRPO group of n samples for a case. If sample i's top-1 answer is a, and k_a samples in the group give a,

$$r_i^{\mathrm{gm}} = \mathrm{clip}\big(n\, p_{\mathrm{ref}}(a) - (k_a - 1),\ 0,\ 1\big), \tag{1}$$

where p_{ref} is the reference differential. This is sample i's marginal contribution to $n \sum_c \min(q_c, p_{\mathrm{ref}}(c))$, the overlap between the group's empirical answer distribution q and the reference. An answer earns credit while its condition is under-represented relative to its reference mass, and none once it is over-represented. In contrast to multi-answer rewards that score a set inside one completion [17], `gm` acts on the distribution of *sampled* answers, the quantity that collapses. Because the group reward fires only when a group contains dissenting samples, we also test a boosted variant, `gm`$^+$: `gm` weight 3 and training temperature 1.2. [TODO: Puri et al. Multi-RLVR and Multi-RLCR baselines under the same settings (running).]

Results. Table 4 reports means over two seeds. All recipes raise top-1 accuracy by 8–10 points. The per-sample recipes (`acc`, `dist`) slightly *reduce* diversity relative to the starting model, the sharpening documented in §4.1. The group reward removes this. `gm` matches the starting model's diversity, and `gm`$^+$ exceeds it while also being the most accurate recipe. Against `acc`, with paired bootstraps over cases averaged over seeds, `gm`$^+$ improves:

- pass@1 by $+1.6$ $[+1.0, +2.3]$ and pass@32 by $+1.7$ $[+1.0, +2.6]$ points;
- distinct answers by $+0.10$ $[+0.07, +0.13]$ and entropy by $+0.034$ $[+0.022, +0.047]$;
- coverage@32 by $+1.5$ $[+1.0, +2.1]$ points, and TV by -0.014 $[-0.019, -0.010]$.

Plain `gm` improves on `acc` in the same direction with smaller margins (e.g. pass@1 $+1.0$ $[+0.6, +1.6]$, entropy $+0.013$ $[+0.005, +0.021]$). Diversity increases monotonically with group-reward pressure, with no accuracy cost. The absolute effects are modest because the starting policy is already collapsed: there is little exploration for any reward to act on. None of the recipes changes rare-disease accuracy or confidence on Phenopacket Store, where a 1.7B model is at the floor. [TODO: 8B: GRPO from Qwen3-8B-Base, a diverse start, with `acc` vs. `gm`, compared with the official Qwen3-8B (running).]

6 Limitations

DDXPlus is synthetic: its reference differential is the posterior of a rule-based engine, and the simulated pathology is not always ranked first in it (it is for "more than 70%" of patients [7, §3.5]), so we treat TV and coverage as distance to a *reference* posterior rather than to clinical truth. Base models fail to produce parsable answers more often. We count these as errors, which biases against base models and makes our gaps conservative. All models so far are Qwen-family 7–8B models evaluated with a single prompt format. [TODO: Llama-3.1-8B family; prompt variants.] Phenopacket cases are drawn from published literature and may be in pretraining data, which we partly address with era splits. We do not evaluate clinical utility with clinicians.

Recipe	pass@1	pass@32	Distinct	Entropy	Cov@32	TV↓
Qwen3-1.7B (untrained)	0.378	0.435	1.41	0.151	0.162	0.850
`acc`	0.458	0.499	1.36	0.142	0.181	0.829
`dist`	0.458	0.502	1.34	0.131	0.181	0.829
`gm`	0.469	0.510	1.40	0.155	0.186	0.825
$\texttt{gm}^+$	**0.474**	**0.516**	**1.46**	**0.176**	**0.196**	**0.815**

Table 4: GRPO on Qwen3-1.7B, evaluated on DDXPlus test (980 cases, 32 samples, T=0.7); means of two seeds. $\texttt{gm}^+$: group-reward weight 3, training temperature 1.2. Per-sample rewards slightly narrow the model; the group reward preserves, then widens, its differential while improving accuracy.

References

[1] Junying Chen, Zhenyang Cai, Ke Ji, Xidong Wang, Wanlong Liu, Rongsheng Wang, and Benyou Wang. Towards medical complex reasoning with LLMs through medical verifiable problems. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 14552–14573, Vienna, Austria, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.findings-acl.751. arXiv version titled "HuatuoGPT-o1, Towards Medical Complex Reasoning with LLMs".

[2] Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, Girish Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, Nick Ryder, Mikhail Pavlov, Alethea Power, Lukasz Kaiser, Mohammad Bavarian, Clemens Winter, Philippe Tillet, Felipe Petroski Such, Dave Cummings, Matthias Plappert, Fotios Chantzis, Elizabeth Barnes, Ariel Herbert-Voss, William Hebgen Guss, Alex Nichol, Alex Paino, Nikolas Tezak, Jie Tang, Igor Babuschkin, Suchir Balaji, Shantanu Jain, William Saunders, Christopher Hesse, Andrew N. Carr, Jan Leike, Josh Achiam, Vedant Misra, Evan Morikawa, Alec Radford, Matthew Knight, Miles Brundage, Mira Murati, Katie Mayer, Peter Welinder, Bob McGrew, Dario Amodei, Sam McCandlish, Ilya Sutskever, and Wojciech Zaremba. Evaluating large language models trained on code. 2021.

[3] Zhoujun Cheng, Shibo Hao, Tianyang Liu, Fan Zhou, Yutao Xie, Feng Yao, Yuexin Bian, Yonghao Zhuang, Nilabjo Dey, Yuheng Zha, Yi Gu, Kun Zhou, Yuqi Wang, Yuan Li, Richard Fan, Jianshu She, Chengqian Gao, Abulhair Saparov, Haonan Li, Taylor W. Killian, Mikhail Yurochkin, Zhengzhong Liu, Eric P. Xing, and Zhiting Hu. Revisiting reinforcement learning for LLM reasoning from a cross-domain perspective. In *Advances in Neural Information Processing Systems (NeurIPS 2025), Datasets and Benchmarks Track*, 2025.

[4] Mehul Damani, Isha Puri, Stewart Slocum, Idan Shenfeld, Leshem Choshen, Yoon Kim, and Jacob Andreas. Beyond binary rewards: Training LMs to reason about their uncertainty. In *International Conference on Learning Representations (ICLR)*, 2026.

[5] Daniel Danis, Michael J. Bamshad, Yasemin Bridges, Andrés Caballero-Oteyza, Pilar Cacheiro, Leigh C. Carmody, Leonardo Chimirri, Jessica X. Chong, Ben Coleman, Raymond Dalgleish, Peter J. Freeman, Adam S. L. Graefe, Tudor Groza, Peter Hansen, Julius O. B. Jacobsen, Adam Klocperk, Maaike Kusters, Markus S. Ladewig, Allison J. Marcello, Teresa Mattina, Christopher J. Mungall, Monica C. Munoz-Torres, Justin T. Reese, Filip Rehburg, Bárbara

C. S. Reis, Catharina Schuetz, Damian Smedley, Timmy Strauss, Jagadish Chandrabose Sundaramurthi, Sylvia Thun, Kyran Wissink, John F. Wagstaff, David Zocche, Melissa A. Haendel, and Peter N. Robinson. A corpus of GA4GH phenopackets: Case-level phenotyping for genomic diagnostics and discovery. *Human Genetics and Genomics Advances*, 6(1):100371, 2025. doi: 10.1016/j.xhgg.2024.100371.

[6] Arsene Fansi Tchango, Rishab Goel, Julien Martel, Zhi Wen, Gaetan Marceau Caron, and Joumana Ghosn. Towards trustworthy automatic diagnosis systems by emulating doctors' reasoning with deep reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS 2022)*, volume 35, 2022.

[7] Arsene Fansi Tchango, Rishab Goel, Zhi Wen, Julien Martel, and Joumana Ghosn. DDXPlus: A new dataset for automatic medical diagnosis. In *Advances in Neural Information Processing Systems (NeurIPS 2022), Datasets and Benchmarks Track*, 2022.

[8] Xiaoke Huang, Juncheng Wu, Hui Liu, Xianfeng Tang, and Yuyin Zhou. m1: Unleash the potential of test-time scaling for medical reasoning with large language models. In *Proceedings of the Fifth Machine Learning for Health Symposium (ML4H 2025)*, volume 297 of *Proceedings of Machine Learning Research*, pages 369–383. PMLR, 2026.

[9] Ahmadreza Jeddi, Kimia Shaban, Negin Baghbanzadeh, Natasha Sharan, Abhishek Moturu, Elham Dolatabadi, and Babak Taati. When does RL help medical VLMs? disentangling vision, SFT, and RL gains. 2026.

[10] Liwei Jiang, Yuanjun Chai, Margaret Li, Mickel Liu, Raymond Fok, Nouha Dziri, Yulia Tsvetkov, Maarten Sap, Alon Albalak, and Yejin Choi. Artificial hivemind: The open-ended homogeneity of language models (and beyond). In *Advances in Neural Information Processing Systems (NeurIPS 2025), Datasets and Benchmarks Track*, 2025.

[11] Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward Grefenstette, and Roberta Raileanu. Understanding the effects of RLHF on LLM generalisation and diversity. In *International Conference on Learning Representations (ICLR)*, 2024.

[12] Jixuan Leng, Chengsong Huang, Banghua Zhu, and Jiaxin Huang. Taming overconfidence in LLMs: Reward calibration in RLHF. In *International Conference on Learning Representations (ICLR)*, 2025.

[13] Seungseop Lim, Gibaeg Kim, Hyunkyung Lee, Wooseok Han, Jean Seo, Jaehyo Yoo, and Eunho Yang. H-DDx: A hierarchical evaluation framework for differential diagnosis. 2025. NeurIPS 2025 Workshop on Generative AI for Health (GenAI4Health).

[14] Che Liu, Haozhe Wang, Jiazhen Pan, Zhongwei Wan, Yong Dai, Fangzhen Lin, Wenjia Bai, Daniel Rueckert, and Rossella Arcucci. Beyond distillation: Pushing the limits of medical LLM reasoning with minimalist rule-based RL. 2025. Introduces the AlphaMed model; NeurIPS 2025 Workshop on Generative AI for Health (GenAI4Health).

[15] Zhiting Mei, Christina Zhang, Tenny Yin, Justin Lidard, Ola Shorinwa, and Anirudha Majumdar. Reasoning about uncertainty: Do reasoning models know when they don't know? In *Findings of the Association for Computational Linguistics: EACL 2026*, pages 3408–3458, Rabat, Morocco, 2026. Association for Computational Linguistics. doi: 10.18653/v1/2026.findings-eacl.178.

[16] Changdae Oh, Qi Zeng, Qi Qi, Andrey Zhmoginov, Deren Lei, Yun He, Hoang Phan, Hangoo Kang, Azalia Mirhoseini, and Sharon Li. Sharpening tax in post-training. 2026.

[17] Isha Puri, Mehul Damani, Idan Shenfeld, Marzyeh Ghassemi, Jacob Andreas, and Yoon Kim. Escaping the mode: Multi-answer reinforcement learning in LMs. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, volume 306 of *Proceedings of Machine Learning Research*, pages 100127–100146. PMLR, 2026. arXiv version titled "Reaching Beyond the Mode: RL for Distributional Reasoning in Language Models".

[18] Peter West and Christopher Potts. Base models beat aligned models at randomness and creativity. In *Conference on Language Modeling (COLM)*, 2025.

[19] Fang Wu, Weihao Xuan, Danny Xing, Ximing Lu, Mingjie Liu, Yi Dong, Zaid Harchaoui, and Yejin Choi. The invisible leash: Why RLVR may or may not escape its origin. 2025.

[20] Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Yang Yue, Shiji Song, and Gao Huang. Does reinforcement learning really incentivize reasoning capacity in LLMs beyond the base model? In *Advances in Neural Information Processing Systems (NeurIPS 2025)*, 2025.

[21] Jiayi Zhang, Simon Yu, Derek Chong, Anthony Sicilia, Michael R. Tomz, Christopher D. Manning, and Weiyan Shi. Verbalized sampling: How to mitigate mode collapse and unlock LLM diversity. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, volume 306 of *Proceedings of Machine Learning Research*, pages 160073–160152. PMLR, 2026.

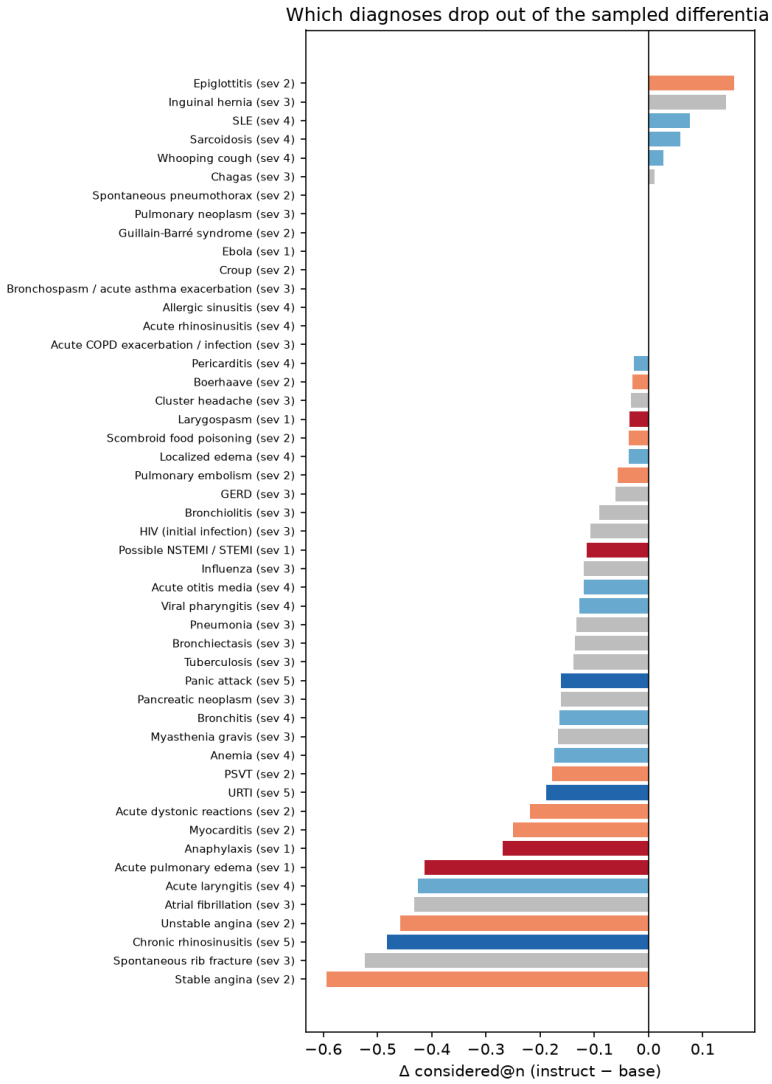


Figure 2: Change in consideration per condition, Qwen2.5-7B-Instruct minus its base model (T=1), colored by severity (red = most severe). Angina, acute pulmonary edema, anaphylaxis and myocarditis are among the largest drops. [TODO: paper-style redraw]

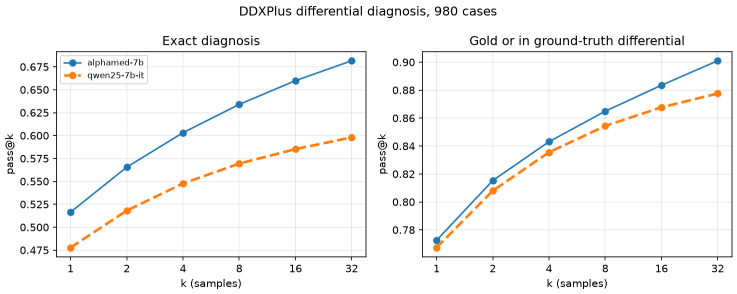


Figure 3: DDXPlus at T=0.7: Qwen2.5-7B-Instruct (dashed) vs. its RL-trained medical descendant AlphaMed. The gap widens with k, so there is no sharpening tax at the medical stage.